Portfolio Energy Performance Report

Scope: Overall portfolio - all 3 buildings (Tech Park Tower A, Grand Residency Hotel, City Square Mall)

Report no.	BCN-20261014-1630-PF	**Report type**	Portfolio
Scope	Overall portfolio - all 3 buildings (Tech Park Tower A, Grand Residency Hotel, City Square Mall)	**Period**	Today, 14 Oct 2026 (00:00 to 16:30)
Generated on	14 Oct 2026, 16:30	**Generated by**	Priya Raman, Portfolio Manager
Data source	BEACON - simulated demo data (15-minute readings)		

1. Executive summary

The 3 buildings used 34,852 kWh of electricity in the period (today), at an energy cost of Rs 2,93,410. The chiller plant was the largest consumer at 35% of the total, running at 0.74 kW/TR against a target of 0.65.

12 savings opportunities were found, worth about Rs 7.24 L per month. The largest is "Chiller plant running at low efficiency" at Tech Park Tower A (Rs 2.15 L per month).

4 faults occurred in the period; 1 is still open. Data completeness was 100.0%.

Metric	Value	Reference / target
Buildings in scope	3	TPA, GRH, CSM
Total floor area	87,000 m2	
Energy consumed	**34,852 kWh**	Today
Energy cost	**Rs 2,93,410**	Grid at time-of-day tariff, DG at Rs 28/kWh
Energy intensity	0.401 kWh/m2	For the period
Energy use intensity (annualised)	210 kWh/m2/yr	Benchmark depends on building type
Peak demand	3,701 kVA	Sum of each building's peak
Solar energy used	2,548 kWh	7.3% of consumption
Chiller plant efficiency	0.74 kW/TR	Target <= 0.65 kW/TR
Avoidable cost in period	Rs 17,535	Energy used with no need, at tariff
Savings opportunities	**Rs 7,24,274 / month**	12 findings, see section on savings
Faults in period	4 (1 open)	
Data completeness	100.0%	201 of 201 readings received

2. Building comparison

Each building's figures for the period. Best value in green, weakest in red.

Metric	Tech Park Tower A	Grand Residency Hotel	City Square Mall
Building type	Office	Hotel	Mall
Floor area (m2)	24,000	18,000	45,000
Energy consumed (kWh)	9,896	10,225	14,730
Energy intensity (kWh/m2)	0.412	0.568	0.327
EUI, annualised (kWh/m2/yr)	216	297	171

Metric	Tech Park Tower A	Grand Residency Hotel	City Square Mall
Energy cost (Rs)	Rs 81,589	Rs 93,766	Rs 1,18,055
Cost per m2 (Rs)	Rs 3.40	Rs 5.21	Rs 2.62
Peak demand (kVA)	1,055	712	1,933
Solar share	7.3%	3.8%	9.8%
Chiller plant (kW/TR)	0.84	0.68	0.71
Savings found (Rs / month)	Rs 4,74,146	Rs 97,956	Rs 1,52,172
Open faults	1	0	0
Data completeness	100.0%	100.0%	100.0%

3. Energy by system

Energy for each system, ranked. Cost is the building cost shared in proportion to energy.

System	Energy (kWh)	Share	Cost est. (Rs)	Key indicator	Target	Status
Chiller Plant	12,078	34.7%	Rs 1,01,681	0.74 kW/TR	<= 0.65 kW/TR	Action needed
Lighting	5,510	15.8%	Rs 46,387	Rs 2,498 avoidable	Zero waste	Review
AHU / FCU	4,946	14.2%	Rs 41,640	Rs 3,017 avoidable	Zero waste	Review
Sockets, IT & Tenants	4,905	14.1%	Rs 41,295	No waste found	Zero waste	Within target
Kitchen, Laundry & Pool	2,182	6.3%	Rs 18,373	No waste found	Zero waste	Within target
Ventilation & Exhaust	1,741	5.0%	Rs 14,656	Rs 2,916 avoidable	Zero waste	Review
STP / WTP	985	2.8%	Rs 8,295	Rs 456 avoidable	Zero waste	Review
Elevators & Escalators	916	2.6%	Rs 7,713	No waste found	Zero waste	Review
Hot Water / Heat Pumps	616	1.8%	Rs 5,183	Rs 785 avoidable	Zero waste	Review
Cold Rooms	525	1.5%	Rs 4,417	Rs 36 avoidable	Zero waste	Review
Water Pumping	356	1.0%	Rs 2,998	No waste found	Zero waste	Within target
UPS Losses	92	0.3%	Rs 772	91.6% efficiency	>= 94%	Review
Total	**34,852**	**100%**	**Rs 2,93,410**			

4. Savings opportunities

Waste found in the readings, ranked by estimated cost per month (based on the last 6 full days).

#	Finding	Bldg	System	Severity	Cost, today (Rs)	Rs / month (est.)	Recommended action
1	**Chiller plant running at low efficiency**	TPA	Chiller Plant	Critical	Rs 7,598	Rs 2,14,954	Rebalance the chilled water valves and reset the secondary pump pressure setpoint.
2	**AHU-06 and AHU-07 running 24/7**	TPA	AHU / FCU	Warning	Rs 1,661	Rs 1,07,640	Put AHU-06 and AHU-07 back on the office-hours schedule.
3	**Car park fans at full speed while the mall is closed**	CSM	Ventilation & Exhaust	Warning	Rs 2,916	Rs 96,660	Enable CO-based speed control on the car park fans.
4	**Chiller plant runs 3 hours after the office closes**	TPA	Chiller Plant	Warning	Rs 0	Rs 69,937	Change the plant stop time to 20:00 on weekdays.

#	Finding	Bldg	System	Severity	Cost, today (Rs)	Rs / month (est.)	Recommended action
5	**AC running in 9 unsold guest rooms**	GRH	AHU / FCU	Warning	Rs 1,355	Rs 60,534	Set unsold rooms to 28 °C setback through the GRMS, linked to the PMS.
6	**Floor 5 lights left on overnight**	TPA	Lighting	Warning	Rs 1,153	Rs 51,030	Link floor 5 lighting to the occupancy sensors and the building schedule.
7	**Facade and signage lights on after sunrise**	CSM	Lighting	Warning	Rs 1,346	Rs 40,365	Fix the photocell / astronomical timer on the facade circuit.
8	**Hot water stored at 68 °C (setpoint 60 °C)**	GRH	Hot Water / Heat Pumps	Info	Rs 785	Rs 35,046	Return the tank setpoint to 60 °C. Do not go below 60 °C.
9	**STP over-aeration**	TPA	STP / WTP	Info	Rs 456	Rs 20,390	Lower the blower DO setpoint to 2.0 mg/L.
10	**Escalators running empty after closing**	CSM	Elevators & Escalators	Info	Rs 0	Rs 15,147	Stop escalators at 22:15, or use slow mode with passenger detection.
11	**UPS running at 18% load (88% efficient)**	TPA	UPS Losses	Info	Rs 228	Rs 10,195	Run one UPS module, or enable eco mode where the IT team allows.
12	**Kitchen cold room door left open**	GRH	Cold Rooms	Info	Rs 36	Rs 2,376	Fit a door alarm and strip curtain on cold room 2.
	Total				**Rs 17,535**	**Rs 7,24,274**	

5. Faults and alarms

Faults that were active at any time in the period.

Fault / alarm	Severity	Bldg	Started	Duration	Status	Acknowledged by
CO2 sensor on floor 3 stuck at 412 ppm	Warning	TPA	13 Oct 2026, 05:30	35 h	Open	-
Gateway offline at City Square Mall	Critical	CSM	14 Oct 2026, 13:15	1 h 15 min	Resolved	-
AHU-03 tripped	Warning	GRH	14 Oct 2026, 14:00	1 h 30 min	Resolved	-
Grid outage, DG-1 running at Grand Residency Hotel	Warning	GRH	14 Oct 2026, 15:00	45 min	Resolved	-

6. Data quality

Readings are expected every 15 minutes from each building's gateway.

Building	Gateway	Readings received	Expected	Completeness	Gaps / notes
Tech Park Tower A	Online	67	67	100.0%	Stuck sensor: CO2 sensor on floor 3 stuck at 412 ppm
Grand Residency Hotel	Online	67	67	100.0%	No gaps
City Square Mall	Online	67	67	100.0%	1 gateway outage

7. Sign-off

Prepared from the energy monitoring system. Review and approve before circulation.

	Name	Designation	Signature	Date
Prepared by	Priya Raman	Portfolio Manager		14 Oct 2026
Reviewed by				
Approved by				